SQLite

- The database that can be used by apps in iOS (and also used by iOS) is called SQLite, and it’s a relational database.
- It is contained in a C-library that is embedded to the app that is about to use it.
- Note that it does not consist of a separate service or daemon running on the background and attached to the app.
- On the contrary, the app runs it as an integral part of it. Nowadays, SQLite lives its third version, so it’s also commonly re ferred as SQLite 3.

Features :

- It’s Open Source.
- It’s lightweight.
- It contains an embedded SQL engine, so almost all of your SQL knowledge can be applied.
- It works as part of the app itself, and it doesn’t require extra active services.
- It’s very reliable.
- It’s fast.
- It’s fully supported by Apple, as it’s used in both iOS and Mac OS.
- It has continuous support by developers in the whole world and new features are always added to it.

The Core SQLite functions

- **sqlite3_open():** This function creates and opens an empty database with the specified filename argument. If the database already exists it will only open the database. Upon return the second argument will contain a handle to the database instance.
- **sqlite3_close():** This function should be used to close a previously opened SQLite database connection. It will free all system resources associated with the database connection.
- **sqlite3_prepare_v2():** To execute an SQL statement it first needs to be compiled into byte-code and that is exactly what this function is doing. It basically transforms an SQL statement written in a string to an executable piece of code.

- **sqlite3_step():** Calling this function will execute a previously prepared SQL statement.
- **sqlite3_finalize():** This function deletes a previously prepared SQL statement from memory.
- **sqlite3_exec():** Combines the functionality of sqlite3_prepare_v2(), sqlite3_step() and sqlite3_finalize() into a single function call.
- **sqlite3_column_<type>():** This routine returns information about a single column of the current result row of a query. Typical values for <type> are text and int. It is important to note that the column indexes are zero based.
- A full list of functions can be found on the SQLite website.